

Bureau of Indian Standards

Manak Bhawan, 9, Bahadur Shah Zafar Marg
New Delhi - 110002

IS 456:2000

2000

PLAIN AND REINFORCED CONCRETE

CODE OF PRACTICE

ICS 91.100.30

BIS 2000

Indian Standard

1 SCOPE

This standard deals with the general construction of plain and reinforced concrete structures. It covers materials, design, and construction requirements.

1 CONCRETE GRADES

Grade	28-day Strength (MPa)	Typical Use
M10	10	Levelling course
M15	15	PCC foundation
M20	20	RCC foundations
M25	25	General RCC
M30	30	Pre-stressed concrete
M35	35	Pre-stressed concrete
M40	40	Pre-stressed concrete

2 MIX PROPORTIONING

The mix proportioning shall be done in accordance with IS 10262. The water-cement ratio shall not exceed the values given in Table 5. Minimum cement content for different exposure conditions:

Exposure Condition	Min Cement (kg/m ³)	Max w/c ratio
Mild	220	0.60

Moderate	240	0.55
Severe	250	0.50
Very Severe	260	0.45
Extreme	280	0.40

3 WORKABILITY

Place of Placement	Slump (mm)	Compacting Factor
Foundations - lightly reinforced	25-75	0.80-0.90
Beams and columns - normal	75-125	0.85-0.92
Thin sections - congested steel	100-150	0.90-0.95

4 DURABILITY REQUIREMENTS

The minimum grade of concrete for reinforced concrete under different exposure conditions shall be:

- Mild exposure: M20
- Moderate exposure: M25
- Severe exposure: M30
- Very severe exposure: M35
- Extreme exposure: M40

5 REINFORCEMENT

Reinforcement shall conform to IS 1786 for high strength deformed bars and IS 432 (Part 1) for mild steel bars. The clear cover to reinforcement shall be as specified in Table 16.

6 QUALITY CONTROL

The strength of concrete shall be determined by compression tests on 150mm cubes at 28 days as per IS 516. For every 50 m³ or part thereof, a minimum of one cube set (3 cubes) shall be tested.